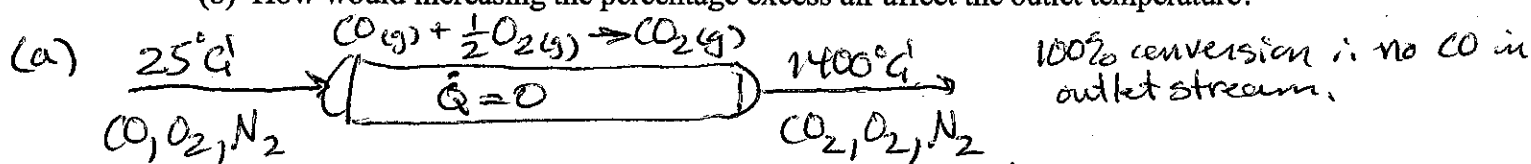


Carbon monoxide (CO) is to be burned with excess air in an adiabatic reactor. Assume 100% conversion of the CO and complete combustion. The reactants are fed at 25°C and the outlet gases are at 1400°C. (a) Calculate the percentage excess air fed to the reactor. (b) How would increasing the percentage excess air affect the outlet temperature?



Material Balance (Basis: $\dot{n}_{\text{CO}, \text{in}} = 1 \text{ mol/s}$)

$$\dot{n}_{\text{CO}, \text{out}} = \dot{n}_{\text{CO}, \text{in}} - \dot{\xi} \Rightarrow \dot{\xi} = \dot{n}_{\text{CO}, \text{in}} = 1 \text{ mol/s}$$

$$\dot{n}_{\text{O}_2, \text{out}} = \dot{n}_{\text{O}_2, \text{in}} - \frac{1}{2}\dot{\xi} \Rightarrow \dot{n}_{\text{O}_2, \text{out}} = \dot{n}_{\text{O}_2, \text{in}} - \frac{1}{2} \text{ mol/s}$$

$$\dot{n}_{\text{CO}_2, \text{out}} = \dot{n}_{\text{CO}_2, \text{in}} + \dot{\xi} \Rightarrow \dot{n}_{\text{CO}_2, \text{out}} = 1 \text{ mol/s}$$

$$\dot{n}_{\text{N}_2, \text{out}} = \dot{n}_{\text{N}_2, \text{in}} \Rightarrow \dot{n}_{\text{N}_2, \text{out}} = \dot{n}_{\text{N}_2, \text{in}} = \left(\frac{0.79 \text{ mol N}_2}{0.21 \text{ mol O}_2}\right) \dot{n}_{\text{O}_2, \text{in}} = 3.76 \dot{n}_{\text{O}_2, \text{in}}$$

Energy Balance (ignore mechanical work and energy)

$$\Delta \dot{H} = \dot{Q} = 0 \text{ (adiabatic reactor)} \quad (\hat{H}_i = 0 @ 25^\circ\text{C})$$

$$\dot{\xi} \Delta \hat{H}_r^\circ + \sum_e \dot{n}_e \hat{H}_e - \sum_i \dot{n}_i \hat{H}_i = \dot{\xi} \Delta \hat{H}_r^\circ + \dot{n}_{\text{CO}_2, \text{out}} \hat{H}_{\text{CO}_2}(1400^\circ\text{C}) + \dot{n}_{\text{O}_2, \text{out}} \hat{H}_{\text{O}_2}(1400^\circ\text{C}) + \dot{n}_{\text{N}_2, \text{out}} \hat{H}_{\text{N}_2}(1400^\circ\text{C}) = 0$$

$$\text{Find } \Delta \hat{H}_r^\circ = \sum_i \nu_i (\Delta \hat{H}_f^\circ)_i = (\Delta \hat{H}_f^\circ)_{\text{CO}_2(\text{g})} - (\Delta \hat{H}_f^\circ)_{\text{CO}(\text{g})} - \frac{1}{2} (\Delta \hat{H}_f^\circ)_{\text{O}_2(\text{g})}$$

$$= [(-393.5) - (-110.53) - \frac{1}{2}(0)] \frac{\text{kJ}}{\text{mol}} = -282.97 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{H}_r^\circ = -\sum_i \nu_i (\Delta \hat{H}_c^\circ)_i = -(\Delta \hat{H}_c^\circ)_{\text{CO}_2(\text{g})} + (\Delta \hat{H}_c^\circ)_{\text{CO}(\text{g})} + \frac{1}{2} (\Delta \hat{H}_c^\circ)_{\text{O}_2(\text{g})}$$

$$= [-(0) + (-283) + \frac{1}{2}(0)] \frac{\text{kJ}}{\text{mol}} = -283 \frac{\text{kJ}}{\text{mol}}$$

Combine Material and Energy Balance

$$\dot{\xi} \Delta \hat{H}_r^\circ + \dot{n}_{\text{CO}_2, \text{out}} \hat{H}_{\text{CO}_2}(1400^\circ\text{C}) + (\dot{n}_{\text{O}_2, \text{in}} - \frac{1}{2} \text{ mol/s}) \hat{H}_{\text{O}_2}(1400^\circ\text{C}) + 3.76 \frac{\text{mol N}_2}{\text{mol O}_2} \dot{n}_{\text{O}_2, \text{in}} \hat{H}_{\text{N}_2}(1400^\circ\text{C}) = 0$$

$$(1 \frac{\text{mol}}{\text{s}})(-283 \frac{\text{kJ}}{\text{mol}}) + (1 \frac{\text{mol}}{\text{s}})(71.89 \frac{\text{kJ}}{\text{mol}}) + (\dot{n}_{\text{O}_2, \text{in}} - \frac{1}{2} \frac{\text{mol}}{\text{s}})(47.07 \frac{\text{kJ}}{\text{mol}}) + 3.76 \dot{n}_{\text{O}_2, \text{in}}(44.51 \frac{\text{kJ}}{\text{mol}}) = 0$$

$$\dot{n}_{\text{O}_2, \text{in}} = 1.094 \text{ mol/s} \Rightarrow \% \text{ excess} = \frac{1.094 \text{ mol/s} - (\frac{1/2}{1})(1 \text{ mol/s})}{(\frac{1/2}{1})(1 \text{ mol/s})} \times 100 = 119\%$$

(b) The outlet temperature would go down.
 The reaction would produce the same amount of energy, but there would be more 25°C material to heat.